

AI for Economic Research: Dynamic Models, Language, and Agents

Lecture 7.4: Economic Applications: Textual Analysis

Zhigang Feng

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Topic 7.4 Agenda

1. **Economic Applications** — FOMC, SEC filings, news
2. **Domain-Specific Models** — FinBERT, CentralBankRoBERTa
3. **Case Studies** — ECB Word2Prices, validation checklist

Arc: T1a Why+Tokens → T1b Embeddings → T2a Attention → T2b Transformer → T3a Training → T3b Limits → T4 Applications.

Economic Applications

Three Featured Case Studies

1. **ECB Word2Prices** (Araujo, Bokan, Comazzi & Lenza, 2025)
 - Word2Vec on ECB statements + BVAR for inflation forecasting
2. **Patents and Worker Exposure** (Kogan et al., 2019)
 - TF-IDF + cosine similarity between patents and occupational tasks
3. **Central Bank Sentiment with BERT** (Pfeifer & Marohl, 2023; Ganguli et al., 2024)
 - Fine-tuned RoBERTa for sentiment toward different economic agents
 - PaECTER patent embeddings for innovation dynamics

After the three deep dives, we briefly tour the wider “application zoo” and the validation pitfalls econometricians need to keep in mind.

Case 1 — ECB Word2Prices: Setup

Araujo, Bokan, Comazzi & Lenza (2025), ECB WP No. 3047

- **Question:** can embeddings of ECB communications improve euro-area inflation forecasting?
- **Data:**
 - ECB press conference introductory statements, 2002Q1–2023Q4
 - Target: HICP_x (core inflation), monthly → quarterly
- **Method:**
 - For each quarter q , retrain Word2Vec on the corpus available *up to* q (no look-ahead)
 - Average word vectors per statement; aggregate to a quarterly text vector
 - Stack text vector with macro variables; fit a Bayesian VAR
 - Forecast 1–4 quarters ahead
- **Why this design?** Re-training quarterly avoids the look-ahead bias that plagues most LLM-based forecast exercises, while remaining computationally cheap.

Case 1 — Python Skeleton

```
# Quarterly Word2Vec retraining for ECB inflation forecasting
from gensim.models import Word2Vec
from statsmodels.tsa.api import VAR
import numpy as np

forecasts = []
for q in quarters:
    # 1. Build corpus available at quarter q (no look-ahead)
    corpus_q = [s for s in all_statements if s.date <= q]
    # 2. Train Word2Vec on the available corpus
    w2v = Word2Vec(corpus_q, vector_size=100, window=5,
                   sg=0, min_count=2, epochs=20)
    # 3. Average word vectors per statement, then per quarter
    stmt_vec = [np.mean([w2v.wv[w] for w in s if w in w2v.wv], axis=0)
                 for s in corpus_q]
    text_q = aggregate_quarterly(stmt_vec, dates)
    # 4. Stack with macro variables and fit a BVAR
    Y = np.column_stack([macro_q, text_q])
    bvar = VAR(Y).fit(maxlags=4)
    # 5. Forecast 1-4 quarters ahead
    forecasts.append(bvar.forecast(Y, steps=4))
```

Case 1 — Results and Lessons

- **Forecast performance:**

- BVAR with Word2Vec text features outperforms a baseline AR / BVAR without text at horizons of 1–4 quarters
- Sentiment dictionaries (e.g. Loughran-McDonald) deliver only marginal gains relative to the no-text baseline
- Heavy LLMs (BERT, fine-tuned) can match or beat Word2Vec but are much more expensive and harder to retrain quarterly — look-ahead bias becomes a real risk

- **Take-aways for empirical economists:**

- “Lighter is better” if it lets you respect the timing structure of the forecasting exercise
 - Frequently-retrained Word2Vec is a strong baseline before reaching for an LLM
 - Always document: training corpus cutoff at every step, vector dimension, window, retraining frequency
- Replicable framework with zero GPU requirements — ideal for grad-student empirical work.

Case 2 — Patent Text and Worker Exposure

Kogan, Papanikolaou, Schmidt, Seegmiller (2019, 2023)

- **Question:** how does exposure to technological innovation affect workers' earnings and employment?
- **Method:**
 - Compute TF-IDF vectors for U.S. patent texts and for O*NET occupational task descriptions
 - Measure cosine similarity between each occupation and each patent
 - Aggregate up: occupation-level technological exposure
- **Findings:**
 - High-exposure occupations experience earnings declines and reduced employment stability
 - Effects are present for both low- and high-wage workers
- **Why this is in our lecture:** a textbook example of how a *simple* method (TF-IDF + cosine) on a *thoughtfully chosen* corpus can answer a substantive economic question. You don't need transformers to make a contribution, but you do need a clean identification strategy.

Why Generic LLMs Underperform on Financial Text

The vocabulary mismatch. Generic BERT/GPT was trained on Wikipedia, web text, and books — not on disclosure language.

- **Polarity flips.** “Liability,” “aggressive,” “exposure” are negative in everyday English but neutral or positive in accounting/finance. Loughran & McDonald (2011) is the canonical correction of the older Harvard-IV-4 list.
- **Hedging language.** “May,” “could,” “intend to” are stop-words in generic NLP but encode legal disclosure stance in 10-K text.
- **Numbers in context.** Subword tokenizers split 1.2345 into digit tokens; without finance fine-tuning the model treats the magnitude as just another vocabulary item.

Two responses: (1) train a domain-specific encoder (next slide), or (2) zero-shot prompt a frontier LLM and validate carefully (Hansen & Kazinnik, 2023).

Case 3a — FinBERT and CentralBankRoBERTa: Two Domain Encoders

	FinBERT (Araci, 2019; Yang et al., 2020)	CentralBankRoBERTa (Pfeifer & Marohl, 2023)
Backbone	BERT-base (12 layers, 110M params)	RoBERTa-base, fine-tuned
Pre-train corpus	Reuters TRC2-Financial + 10-K corpora	RoBERTa-base fine-tuned on ~12k hand-labeled sentences (plus ECB/BIS pseudo-labels)
Task it ships for	3-way sentiment (positive/negative/neutral) on financial sentences	Agent-specific sentiment: stance toward five macro agents (households, firms, financial sector, government, central bank)
Headline result	Beats LSTM/dictionary baselines on Financial PhraseBank	Agent-specific scores predict regional inflation & employment better than “hawkish/dovish”
When to reach for it	Fast firm-level sentiment at scale	Central-bank communication studies

Both are fine-tuned BERTs — the architecture is from Topic 7.2; the pre-training/fine-tuning recipe is from Topic 7.3a. T4 cares only about *using* them: load weights, feed sentences, get scores.

Domain Models vs. Frontier LLMs: No Single Winner

Hansen & Kazinnik (2023): “GPT-4 reads Fedspeak.” Zero-shot GPT-4 classifies FOMC sentence stance at expert-level accuracy; beats fine-tuned BERT/RoBERTa; provides human-consistent explanations.

Gambacorta et al. (BIS WP 1215, 2024): CB-LMs. Domain-adapted encoder beats generic BERT on FOMC stance; GPT-4 still wins on complex narrative tasks.

Practical decision rule:

- **Sentence-level sentiment at scale, fixed taxonomy** → FinBERT / CentralBankRoBERTa (cheap, deterministic, citable).
- **Open-ended interpretation, narrative, novel category** → frontier LLM (GPT-4, Claude) with a logged prompt.
- **In doubt** → run both and report agreement (Cohen's κ).

No single model dominates — the validation checklist (later in this topic) handles the choice.

Case 3b — PaECTER for Patent Innovation

Ganguli, Lin & Reynolds (2024), NBER WP

- **Goal:** measure innovation dynamics from patent text using transformer-based embeddings.
- **Model: PaECTER** (Patent Embeddings from Citation-Enhanced Transformer Representations)
 - Initialize from BERT-base
 - Fine-tune on patent corpora
 - Use citation pairs as a supervised signal during fine-tuning (related patents pulled together in embedding space)
- **Findings:**
 - Significantly more accurate similarity judgments than TF-IDF or LDA
 - Distinguishes incremental from radical innovations
 - Long-run trend: patents are increasingly diverging in semantic space (broader exploration)
- **Take-away:** fine-tuned transformer embeddings give a richer, citation-aware view of technological evolution than purely text-based or count-based methods.

Frontier (2024–25): LLMs for Central Bank Communication

- **CB-LMs** (Gambacorta et al., BIS WP 1215, 2024)
 - Domain-adapted encoder retrained on 37,000 central bank papers + 18,000 speeches
 - Outperforms generic BERT on FOMC stance classification
 - But: GPT-4 wins on complex narrative tasks — no single model dominates
- **GPT-4 for FedSpeak** (Hansen & Kazinnik, 2023)
 - GPT-4 classifies FOMC statement sentences at *expert-level* accuracy
 - Outperforms fine-tuned BERT/RoBERTa; provides human-consistent explanations
 - Zero-shot performance collapses domain-adaptation barrier for many tasks
- **GPT-4o on 25,000 FOMC Sentences** (Federal Reserve FEDS Note, December 2024)
 - Topic classification across 2010–2024 minutes at scale
 - Detects “financial stability” discourse spikes in 2020 and 2023 banking crises
 - LLMs reveal *thematic shifts* that sentiment scores miss
- **74,882 Central Bank Documents** (Silva, Moriya & Veyrune, IMF WP 2025/109)
 - 169 central banks, 1884–2025; LLM-coded stance + directional communication index
 - Forecasts future policy rate changes; documents shift to forward-looking language

Frontier (2024–25): LLMs in Asset Pricing

- **Expected Returns and LLMs** (Chen, Kelly & Xiu, NBER WP 31502, 2024)
 - LLM embeddings of financial news $\Rightarrow$ cross-sectional equity return prediction
 - Substantially outperforms TF-IDF and momentum; gains largest on *negation* and complex narratives
 - Evidence of slow price adjustment to information that only LLM embeddings extract
- **AI Asset Pricing Models** (Kelly, Kuznetsov, Malamud & Xu, NBER WP 33351, 2025)
 - Transformer architecture embedded directly in the stochastic discount factor M_{t+1}
 - Multi-head attention learns cross-asset dependencies from 132 stock characteristics
 - Large reductions in pricing errors vs. previous ML factor models
- **Asset Embeddings** (Gabaix, Koijen, Richmond & Yogo, NBER WP 33651, 2025)
 - Embeddings learned from *investor portfolio holdings* via skip-gram (same math as Word2Vec)
 - Arithmetic: `tech + growth - hardware` $\approx$ `software firms`
 - GPT-4 generates economic narratives for firm clusters in the learned embedding space
- **Key insight:** the same transformer and embedding toolkit that encodes language can encode financial structure — text and quantitative data are no longer separate worlds.

The Application Zoo, Organized by Econometric Role

The same text can play three very different roles. Pick the representation to match the role.

Paper	Signal (regressor $m(x_i)$)	Outcome (behavior y)	Instrument / shock
Baker–Bloom–Davis (2016)	Newspaper keyword EPU $\rightarrow$ VAR	—	EPU surprises
Hoberg & Phillips (2016)	10-K TF-IDF cosine $\rightarrow$ product-market similarity	—	—
Hassan et al. (2019)	Bigram firm-level political risk $\rightarrow$ investment regression	—	Quarterly call shocks
Li et al. (2021)	W2V on earnings calls $\rightarrow$ 5 culture dimensions	—	—
Lopez-Lira & Tang (2023)	ChatGPT zero-shot sentiment on headlines $\rightarrow$ next-day returns	—	—
Hansen–McMahon–Prat (2018)	LDA topics	Committee speech change after transparency reform	1993 reform date
Hartley (2025)	LLM-reconstructed EPU (19th c., OCR)	—	War-news shocks

- *Signal column dominates the literature* — it's the easiest role and the loudest result.
- *Outcome* requires a behavior to be *caused* by something else; harder but cleaner identification.
- *Instrument / shock* use needs the exclusion restriction to survive the prompt/fine-tuning recipe — document it.^{14 / 19}

Validation Caveats for Econometricians

- **Cosine similarity is not innocent.** Three common confounds inflate similarity between firm/policy texts:
 - **Industry jargon:** firms in the same SIC code share boilerplate language
 - **Templates:** regulatory filings re-use chunks of text verbatim
 - **Length:** longer documents have smoother averages and look more similar
- **Standard tools for robustness:**
 - Permutation tests — shuffle labels within an industry, check if observed similarity exceeds the within-industry baseline
 - Residual embeddings — subtract the industry mean before measuring similarity
 - Bootstrap the entire pipeline (text → embedding → measurement → regression) for honest standard errors
- **Dictionary methods are not innocent either.** The Loughran-McDonald (2011) finance-specific dictionary corrects mismatches in the older Harvard IV-4 list (e.g. “liability” is not negative in financial accounting).
- **Always validate** a sample of model outputs against human coding before scaling up.

A Six-Step Workflow for LLM-Assisted Measurement

1. **Pre-alignment.** Hand-code 100–200 examples; iterate on the prompt or fine-tuning recipe until you reach acceptable agreement.
2. **Full annotation.** Run the model on the entire sample with a frozen prompt and a frozen model version.
3. **Gold-standard audit.** Blind-code 10% of the sample and compute Cohen's κ (publication threshold $\kappa \geq 0.75$).
4. **Category-level reporting.** Don't just report overall accuracy; show the confusion matrix by class. Minority classes are often where models fail silently.
5. **External validity check.** Link the text-derived measure to a real economic outcome (returns, investment, employment) before treating it as a measurement.
6. **Disclosure.** Report prompt, model version, temperature ($\tau = 0$), date, all preprocessing, and provide code. Reproducibility is essential when the model behind the API can change without notice.

Worked Example: Applying the Checklist to FOMC Sentiment

Imagined paper: “GPT-4 sentiment on FOMC statements predicts 10-year yields.”

Step	What you do	What goes in the appendix
1. Pre-align	Hand-code 200 sentences hawk/dove/neutral	Prompt v3 (frozen); κ vs. author = 0.81
2. Annotate	GPT-4-2024-06-13, $\tau = 0$, full corpus	Cost, runtime, retries, sha256 of inputs
3. Gold audit	Blind-code 10% by RA	$\kappa = 0.78$ overall; 0.62 on “neutral”
4. By-class	Confusion matrix	Hawkish recall 0.91; neutral recall 0.66 (flagged)
5. External validity	Predict Δy_{10} at FOMC dates	$\hat{\beta} = 4.2$ bp per unit, $t = 3.1$
6. Disclose	Prompt + code on Zenodo	DOI, model card, replication run-time

LLM-as-judge connection. A second LLM as a cross-checker sub-agent: same input, different prompt, flag disagreements for human review — the same construction-to-coefficient logic used in AI-exposure scoring (Topic 7.3b).

Disclosure Norms for AI-Assisted Empirical Work

Minimum reproducibility package.

- **Model identity.** Provider, model name, exact version string, call date.
- **Prompt(s).** Frozen text in a versioned file; not paraphrased in the paper.
- **Sampling parameters.** Temperature, top- p , max tokens, seed if available.
- **Inputs.** sha256 of every input file; document any preprocessing.
- **Outputs.** Raw model outputs preserved *before* any post-processing.
- **Code.** End-to-end script with retry logic and the exact API client version.

AEA policy. The October 2024 Data and Code Availability update explicitly covers AI-assisted analysis: report when an LLM was used, for what step, and provide artifacts that let a reader reproduce the result given the same model.

IRB and proprietary text. Survey transcripts, internal memos, and clinical text often cannot be sent to a hosted API. Run an open-weights model locally, or obtain IRB sign-off on the provider's data-handling terms.

The closed-source LLM is the most-changed equipment in your empirical paper. Document it like the dataset it is.

Topic 7.4 Wrap-Up — and the Bridge to Lec08/Lec09/Lec10

- **Three econometric roles of text.** Signal (regressor), outcome (behavior), instrument (shock). The role determines the representation: transparent for measurement, contextual for behavior, frozen-snapshot for instruments.
- **Six-step measurement workflow.** Pre-align → full annotation → gold-standard audit → category-level reporting → external validity → disclosure. Cohen's $\kappa \geq 0.75$ is the publication bar.
- **Disclosure norms (AEA, October 2024).** Model identity, prompt, sampling parameters, input hashes, raw outputs, end-to-end code. Treat the LLM call as you would a survey instrument.
- **What's next.**
 - **Lec08 (RAG)** — ground the LLM in your private corpus. Solves T3b's hallucination + cutoff + context-window limits via retrieval; the token-vs-doc-embedding frame from T2b is the geometric prerequisite.
 - **Lec09 (Agentic AI)** — ground the LLM in actions. JSON schemas from T3a become tool calls; ICL-as-Bayesian (T3a) returns as the context dilemma.
 - **Lec10 (Case Studies)** — end-to-end empirical papers built with the Lec07–Lec09 toolkit.

Language engine (Lec07) + grounded memory (Lec08) + grounded action (Lec09) = auditable economic measurement.